GATE ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 10



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic Number System

Topics to be Covered



Topic-1 Counting Theory

Topic-2 Questionnaire

[MCQ]

#Q. Which largest number of 5 digits is divisible by 99 ?

Assignment

$99 = 9 \times 11$

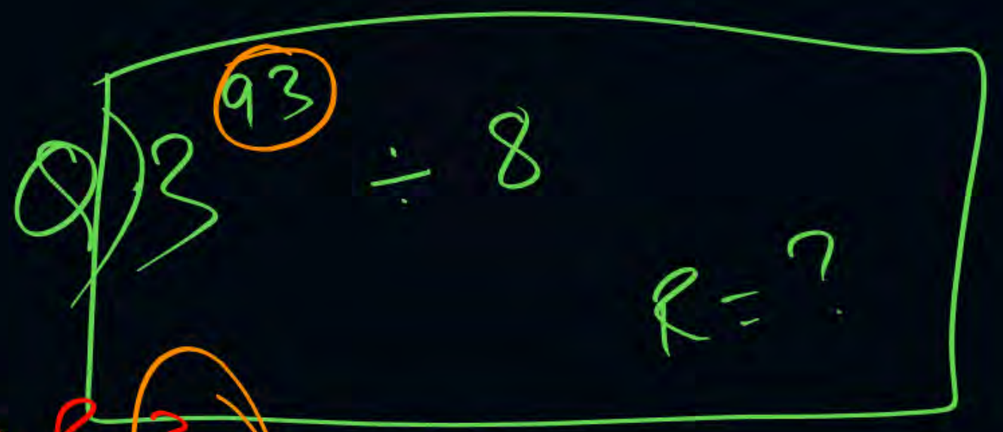
A. 99999 — $9+9+9+9+9 = 45$, $4+5 = 9$

B. 99981 — $9+9 = 18$, $1+8 = 9$

C. 99909 — $9+9+9 = 27$, $2+7 = 9$

D. 99990 — $9+9+9+0 = 27$, $2+7 = 9$

Numbers



$8\overline{)93}$ (11)
3
10
3

1. Count
2. Measurement
3. Comparison
4. Ratio
5. etc

$3^1 = 3 \div 8 \Rightarrow R = 3$

$3^2 = 9 \div 8 \Rightarrow R = 1$

$3^3 = 27 \div 8 \Rightarrow R = 3$

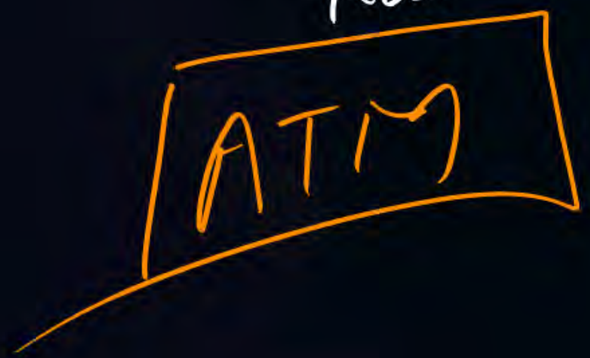
$3^4 = 81 \div 8 \Rightarrow R = 1$

$R = 3$

Counting Theory

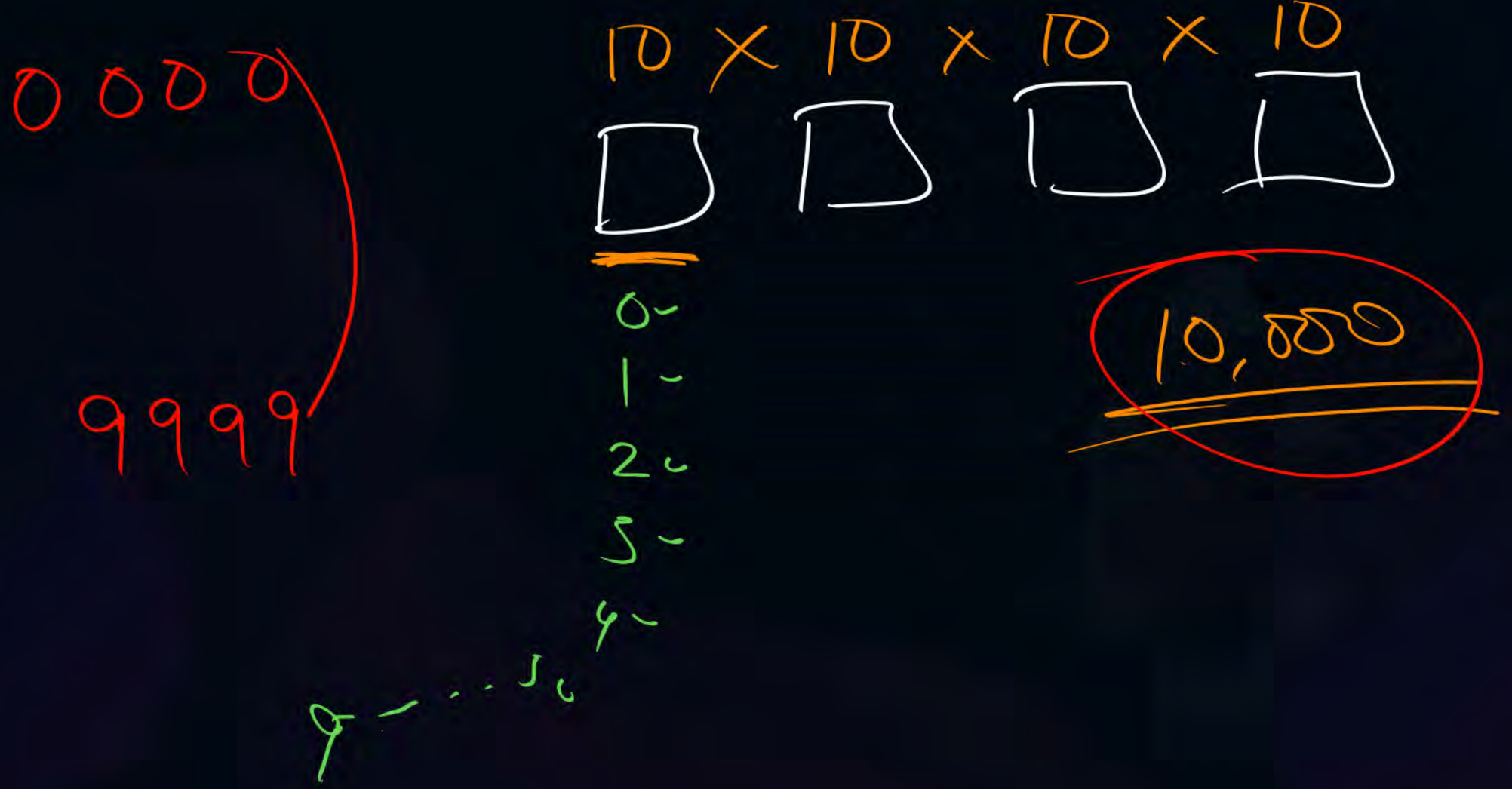


Counting the ways to do a work:



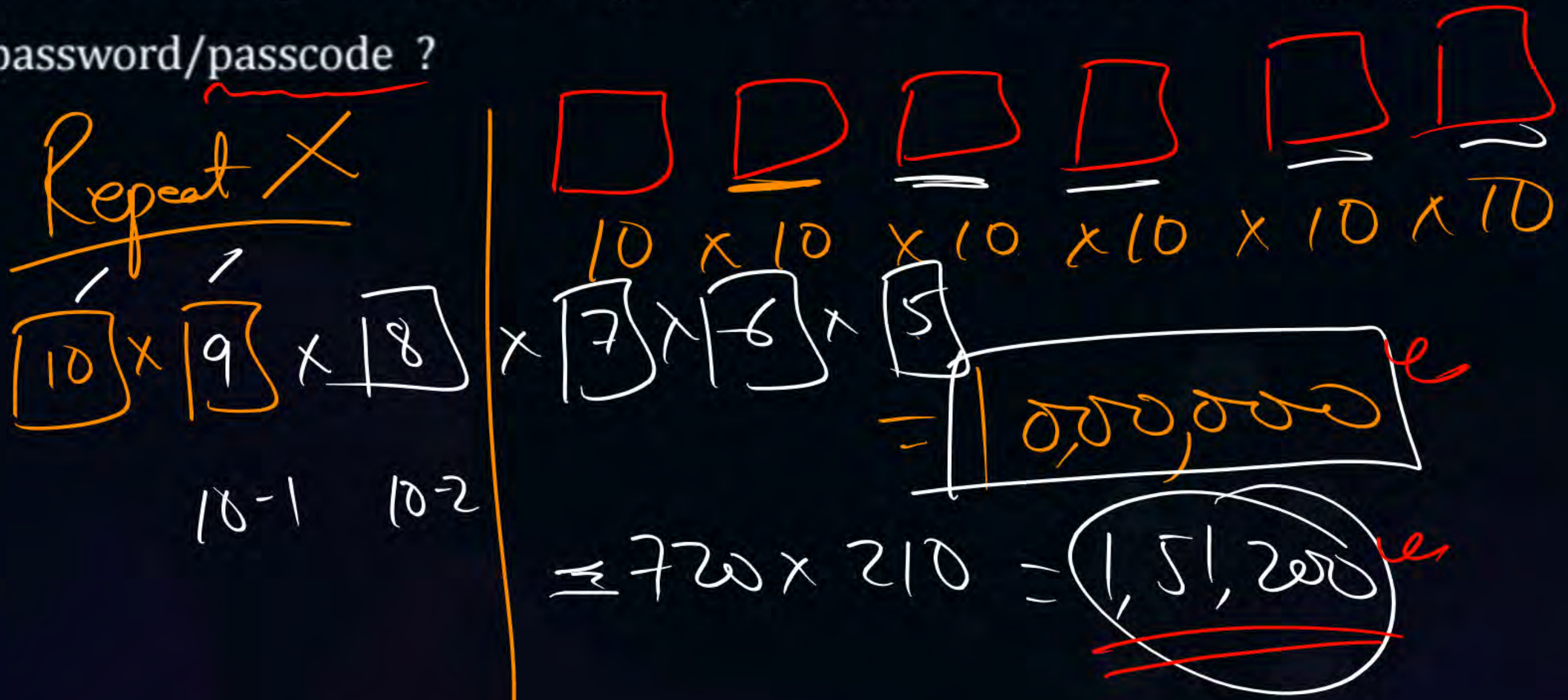
[MCQ]

#Q. In how many different ways you can choose your ATM PIN ?



[MCQ]

#Q. In how many different ways you can select a six digit phone password/passcode ?



[MCQ]

#Q. How many four digits number can be formed using 1, 2, 3, 4 & 5?

Repeat

Th H I U

5 × 5 × 5 × 5

= 625

Repetition

Th H I U

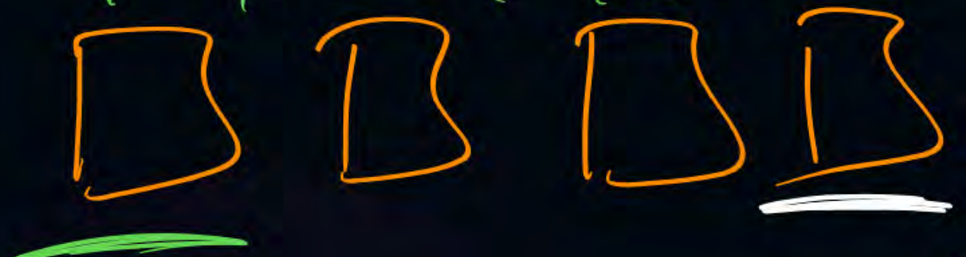
5 × 4 × 3 × 2

= 120

[MCQ]

#Q. In how many different ways you can choose your ATM PIN which is an even ?

$10 \times 10 \times 10 \times 5$

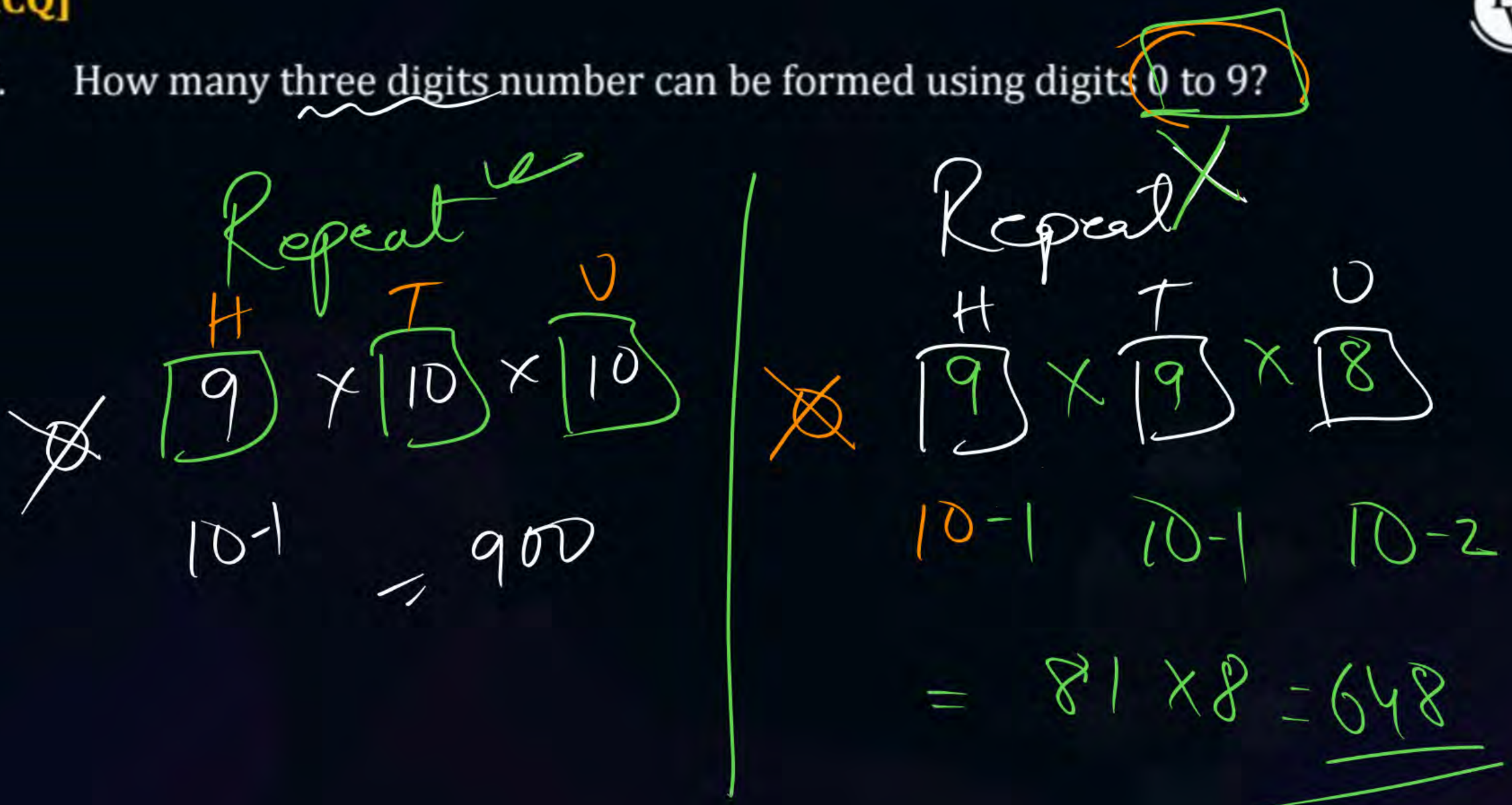


$= 5000$

0
2
4
6
8

[MCQ]

#Q. How many three digits number can be formed using digits 0 to 9?



[MCQ]

#Q. In how many ways four letters can be posted in 6 post boxes, if each box can take any number letters?

A 4096

B 1500

C 24

D 1296

$L_1 \quad L_2 \quad L_3 \quad L_4$

$\frac{}{6} \times \frac{}{6} \times \frac{}{6} \times \frac{}{6}$

$= 216 \times 6$

$= 1296$

[MCQ]

#Q. In how many ways can you arrange 5 Novels on a shelf?

A 720

B 120

C 360

D 300

$5 \times 4 \times 3 \times 2 \times 1$

$= 120$

[MCQ]

PYQ

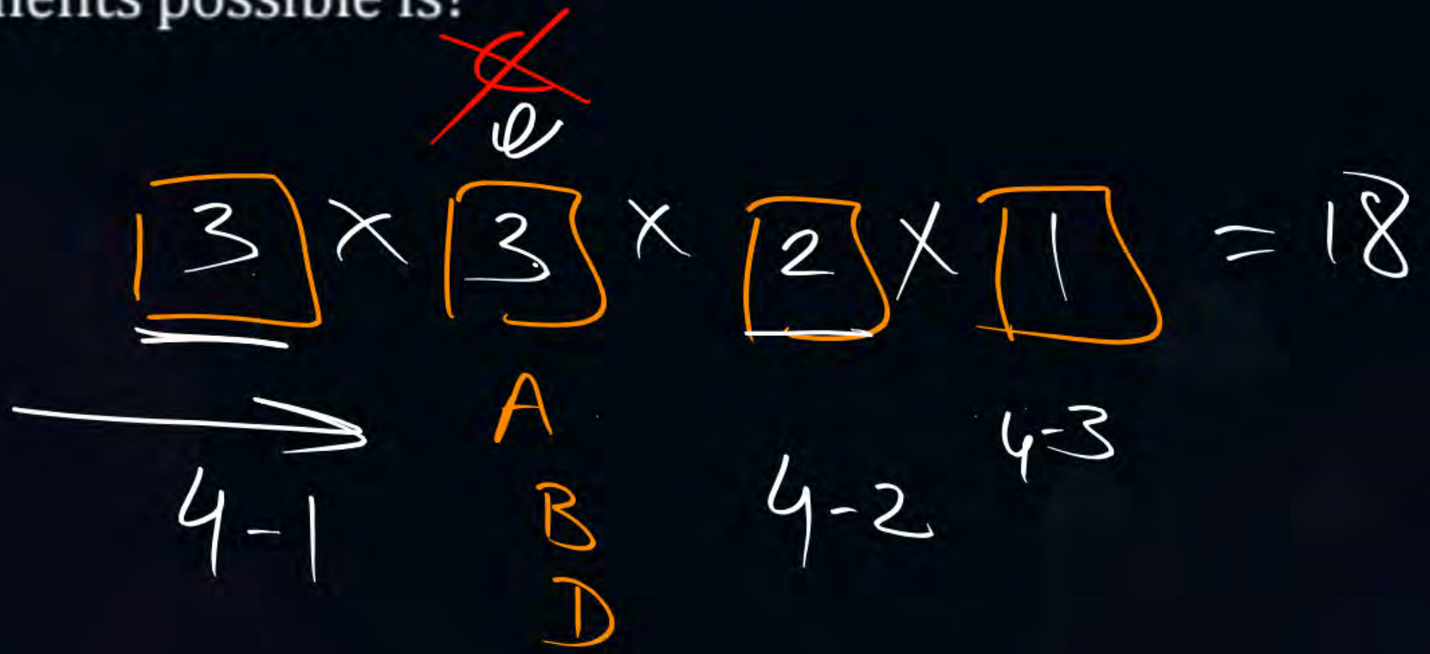
#Q. Four persons A, B, C and D are to be seated in a row. C should not be seated at the second position from the left end of the row. The number of distinct seating arrangements possible is?

A 6

B 9

C 18

D 24



[MCQ]

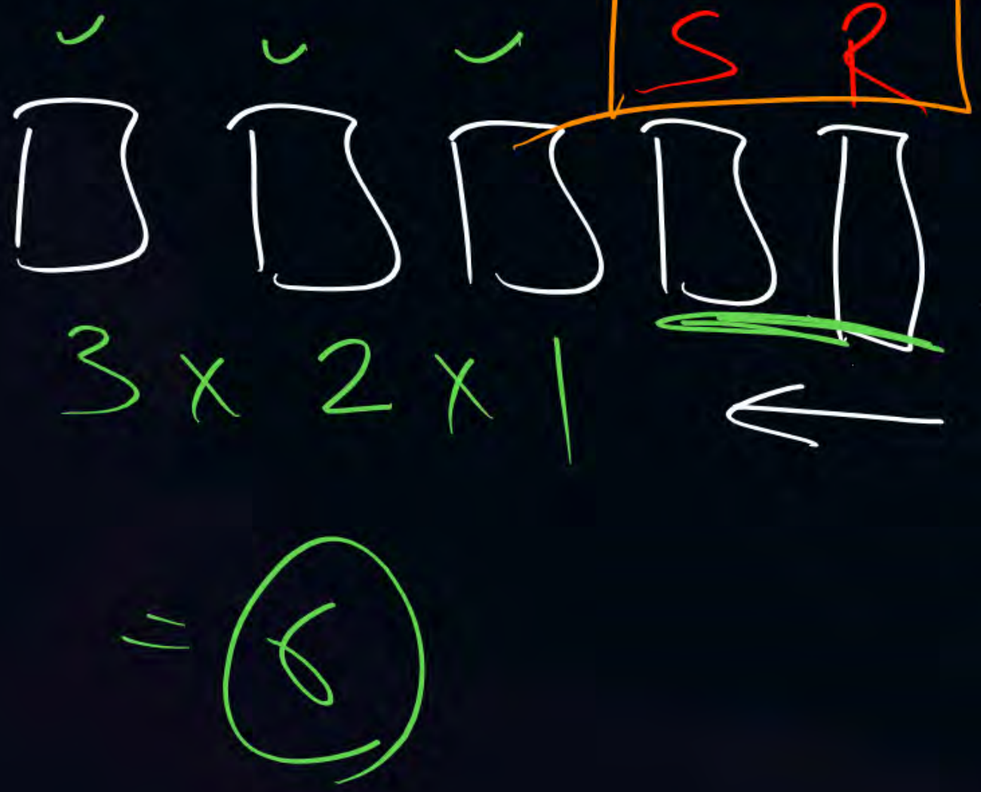
#Q. Five different book (P, Q, R, S, T) are to be arranged on a shelf. The books R and S are to be arranged first and second, respectively from the right side of the shelf. The number of different orders in which P, Q and T may be arranged is?

A 12

B 120

C 6

D 2



[MCQ]

#Q. A license plate begins with 3 letters. If the possible letters are A, B, C, D and E, how many different ways these letters can be written if no letter is used more than once?

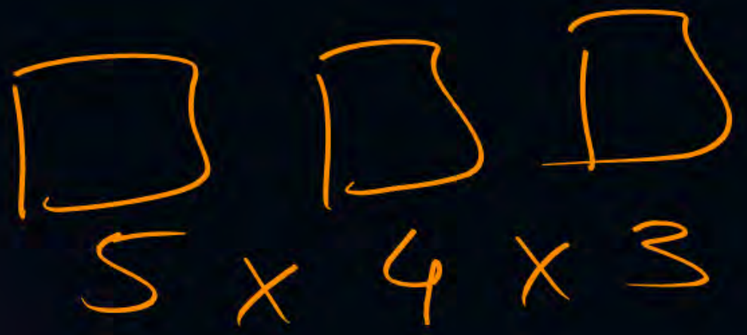
A. 720

B. 120

C. 360

D. 300

60

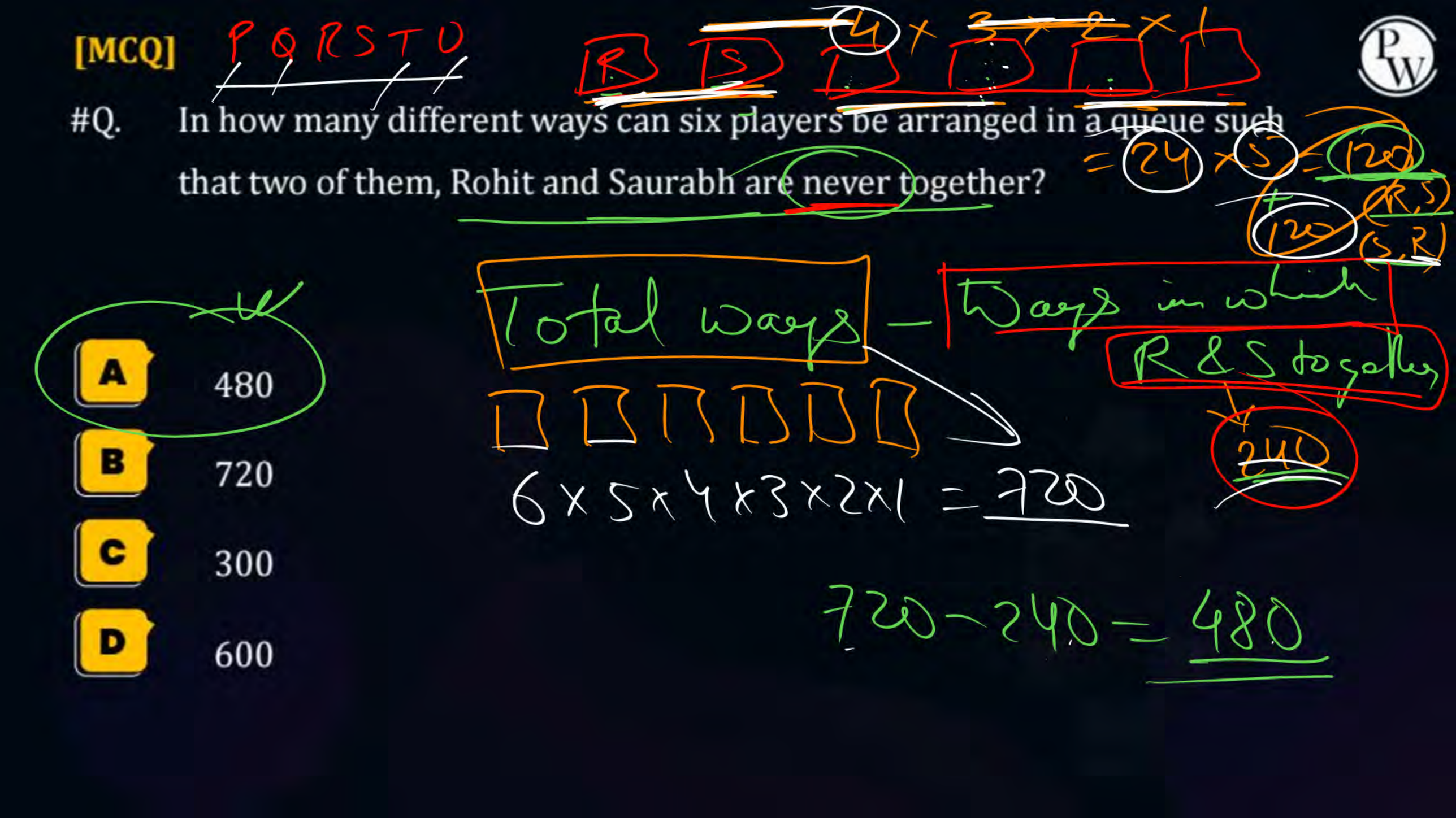


= 60

[MCQ]

#Q. In how many different ways can six players be arranged in a queue such that two of them, Rohit and Saurabh are never together?

A. 480
B. 720
C. 300
D. 600





2 mins Summary

Topic Counting Theory

THANK - YOU